

08/11/2014(E)

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)
Semester: July 2014 - November 2014

Max. Marks: 100**Duration: 3 Hrs****End Semester Exam**

Class: TE

Semester: V

Name of the Course: Operating System

Branch: Computer Engineering

Course Code: UCEC502

Instructions:

- (1) All Questions are Compulsory.
- (2) Draw suitable diagram wherever applicable.

Question No.		Max. Marks																		
Q1 (a)	Explain the operating system structures with the help of suitable diagram. OR What is an operating system? Describe the services provided by operating systems?	10																		
Q1 (b)	What is process with the help of suitable diagram? Describe the operation of process creation and termination in UNIX.	10																		
Q2	Attempt any Two questions out of remaining three questions.																			
(a)	Consider the following set of processes, with the length of the CPU burst given in milliseconds: <table border="1"> <thead> <tr> <th>Process</th><th>Burst Time</th><th>Priority</th></tr> </thead> <tbody> <tr> <td>P₁</td><td>15</td><td>3</td></tr> <tr> <td>P₂</td><td>2</td><td>1</td></tr> <tr> <td>P₃</td><td>4</td><td>3</td></tr> <tr> <td>P₄</td><td>2</td><td>4</td></tr> <tr> <td>P₅</td><td>8</td><td>2</td></tr> </tbody> </table> The processes are assumed to have arrived in the order P₁, P₂, P₃, P₄, and P₅ all at time 0. i) Draw Gantt Charts to show execution using First Come First Served (FCFS), Non-preemptive Shortest Job First (SJF), Non-preemptive Priority (a smaller priority number implies a higher priority) ii) What is waiting time and turnaround time for each process? iii) Which of the above scheduling algorithms results in minimal average waiting time?	Process	Burst Time	Priority	P ₁	15	3	P ₂	2	1	P ₃	4	3	P ₄	2	4	P ₅	8	2	10
Process	Burst Time	Priority																		
P ₁	15	3																		
P ₂	2	1																		
P ₃	4	3																		
P ₄	2	4																		
P ₅	8	2																		
(b)	What is meant by critical section problem? Give its general structure. Explain the requirements that must be satisfied by a solution to the critical section problem.	10																		
(c)	Explain I/O Hardware – Polling, Direct Memory Access (DMA), Interrupt.	10																		

Pg 1/2

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Q3 (a)	What are semaphores? Explain the solution to producer-consumer problem using semaphores.	10																																																																					
Q3 (b)	What are the necessary and sufficient conditions to occur deadlock? Explain the resource allocation graph scheme to deal with deadlock problem? What are the limitations of this approach? OR Consider the following snapshot of a system: <table><tr><th rowspan="2">Process</th><th colspan="3">Allocation</th><th colspan="3">Max,</th><th colspan="3">Available</th></tr><tr><th>A</th><th>B</th><th>C</th><th>A</th><th>B</th><th>C</th><th>A</th><th>B</th><th>C</th></tr><tr><td>P₀</td><td>0</td><td>1</td><td>0</td><td>7</td><td>5</td><td>3</td><td>3</td><td>3</td><td>2</td></tr><tr><td>P₁</td><td>2</td><td>0</td><td>0</td><td>3</td><td>2</td><td>2</td><td></td><td></td><td></td></tr><tr><td>P₂</td><td>3</td><td>0</td><td>2</td><td>9</td><td>0</td><td>2</td><td></td><td></td><td></td></tr><tr><td>P₃</td><td>2</td><td>1</td><td>1</td><td>2</td><td>2</td><td>2</td><td></td><td></td><td></td></tr><tr><td>P₄</td><td>0</td><td>0</td><td>2</td><td>4</td><td>3</td><td>3</td><td></td><td></td><td></td></tr></table> Answer the following question using the Banker's Algorithm - What is the content of matrix "Need"? - Is the system in a safe-state? If so give the safe sequence. - If a request from a process P₄ for (3, 3, 0) arrives, can it be granted immediately?	Process	Allocation			Max,			Available			A	B	C	A	B	C	A	B	C	P ₀	0	1	0	7	5	3	3	3	2	P ₁	2	0	0	3	2	2				P ₂	3	0	2	9	0	2				P ₃	2	1	1	2	2	2				P ₄	0	0	2	4	3	3				10
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P ₄	0	0	2	4	3	3																																																																	
Q4 (a)	What is segmentation? Describe the hardware support required for segmentation with the help of suitable example. OR Describe three different structure of the page table with suitable example.	10																																																																					
Q4 (b)	A drive has 5000 cylinders numbered 0 to 4999. The requested tracks in the order received are 86, 1470, 913, 1774, 948, 1509, 1022, 1750, 130 Apply the following disk scheduling algorithm starting track at 143. Illustrate with figures in each case & calculate the total distance travelled by the disk arm to satisfy the requests using the following disk scheduling algorithms: - First Come-First Serve (FCFS) - Shortest Seek Time First (SSTF) - C-LOOK	10																																																																					
Q5 (a)	What are different types of files in UNIX? List important UNIX Directories & files. Explain Internal structure of UNIX file. OR Explain design principles and component of LINUX.	10																																																																					
Q5 (b)	Explain about contiguous memory allocation.	10																																																																					